Project Proposal: GeoResilient

An Open GIS Framework for Assessing the Structural Resilience of Urban Road Networks

Case Study: Hyderabad, India

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1. Overview

Urban road networks appear well-connected on a map, but in practice, connectivity is often concentrated in a small number of critical links — key bridges, flyovers, or arterial corridors. When these links fail (due to flooding, construction, accidents, or disasters), the practical accessibility of a city can degrade far more than its visual density suggests.

This project models Hyderabad's road network as a graph and quantifies how resilient it is to disruption — identifying which roads and junctions the city depends on most, and measuring how accessibility to essential services (hospitals, schools, fire stations) changes when those links are removed.

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2. Objectives

- Identify the road segments and junctions that the city's connectivity most depends on

- Quantify overall network redundancy, and compare it across different zones of the city
- Measure baseline accessibility to essential services from every neighborhood
- Simulate disruption scenarios (targeted failure, random failure, flood-linked failure) and quantify their impact
- Produce a set of maps and metrics that communicate which parts of Hyderabad are most vulnerable to network breakdown

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3. What Can Be Done (Scope of Analysis)

- Identify the most critical roads in the network
- Identify single points of failure — junctions whose removal disconnects parts of the city
- Compute a redundancy/meshedness score for the network, city-wide and by zone
- Map baseline travel-time accessibility to hospitals, schools, and fire stations
- Simulate targeted removal of the most critical roads and recompute accessibility
- Simulate random road removal for comparison (tests whether the network is "robust yet fragile")
- Simulate flood-linked road removal using terrain/low-lying area data, for a locally grounded disruption scenario
- Compare resilience across zones (e.g., old city vs. HITEC City vs. outskirts)
- Compare redundancy by road class (arterial vs. residential)

4. What Can Be Shown (Expected Outputs)

- Map of the most critical roads in Hyderabad's network
- Map of single-point-of-failure junctions
- Before/after accessibility maps under each disruption scenario
- Ranked list of the most vulnerable neighborhoods/wards
- Numeric resilience/redundancy scores, compared across city zones
- Quantified impact of disruption (e.g., average travel time to nearest hospital before vs. after)
- Population-weighted impact estimates (e.g., number of people who lose fast access to a hospital under a given scenario)
- Evidence for or against the "robust yet fragile" pattern seen in other urban networks worldwide
- A flood-risk-linked vulnerability map, combining terrain exposure with network fragility

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5. How It Can Be Done (High-Level Approach)

1. **Build the network** — extract Hyderabad's road network as a graph (junctions as nodes, roads as edges) from OpenStreetMap

2. **Locate essential services** — extract hospital, school, and fire-station locations and attach them to the network
3. **Score the network** — apply standard graph-theory measures (centrality, connectivity, redundancy indices) to identify critical links and compute an overall resilience baseline
4. **Measure baseline accessibility** — compute travel time from every neighborhood to its nearest essential service
5. **Simulate disruptions** — remove critical roads (and, separately, random roads and flood-prone roads) and recompute accessibility
6. **Compare and visualize** — quantify the accessibility loss under each scenario and map the results

(Full technical methodology — specific algorithms, formulas, and code — to be detailed separately once the project is confirmed.)

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6. Datasets and Sources

Dataset	Purpose	Source
Road network	Core graph (nodes/edges)	OpenStreetMap, via OSMnx / Geofabrik India extracts
Hospitals, schools, fire stations	Essential-service destinations	OpenStreetMap (OSMnx `features_from_place`); cross-check with

Dataset	Purpose	Source
		data.gov.in / Telangana Open Data Portal
Population	Population-weighted impact analysis	WorldPop (100m grid)
Admin/ward boundaries	Zone-wise comparison	datameet/maps; india-geodata
Transit (GTFS)	Optional — multi-modal accessibility	Telangana Open Data Portal (Hyderabad Metro + TGSRTC GTFS); fallback: HMRL Open Data
DEM (elevation/terrain)	Flood-scenario disruption modeling	OpenTopography / USGS EarthExplorer (SRTM 30m); higher-res alternative: Bhuvan Cartosat DEM

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7. Tools & Stack

Python — OSMnx, NetworkX, GeoPandas, Rasterio, Matplotlib/Folium. No GPU or specialized compute required; all datasets are freely available without cost.

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8. Optional Extensions

- Compare Hyderabad against a second Indian city to test whether the vulnerability pattern generalizes
- Analyze resilience separately by service type (does losing a road affect hospital-access differently than school-access?)
- Incorporate time-of-day traffic variation, if speed/congestion data can be sourced
- Extend the flood-disruption scenario using historical flood-affected-area records instead of elevation alone

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9. Relevance / Themes

This project aligns with the following course/CFP themes: *spatial modelling; prediction and simulation; location-based services (LBS); crowdsourcing and VGI; cognitive-based GIS and cartography.*